Oral script: the 10-minute defence

Museum Guide project, MSA-DATB09-03. One section per slide, to read aloud.

How to use it

- About 1256 words: at a calm pace (about 130 words per minute), this is about 10 minutes. Without the 47 optional words, about 9 minutes. The rest of the 10 minutes is a safety margin for pauses, clicks and nerves.
- Every slide ends with a green TRANSITION box, always built the same way: "We saw that... So now...". Say it while you click to the next slide, and never skip it, even when you are late: it turns the six steps into one story.
- Read it aloud 3 times, with the slides, and time yourself once. Speak slowly; pause after each bold sentence: it is the one the jury must remember.
- Blue text in brackets is a stage cue (where to point); do not read it. Grey sentences marked (optional) can be skipped if you are late.
- If you present with a teammate, a natural split is slides 1 to 6, then slides 7 to 12.

The story, transition by transition

1 to 2: To answer it, we did not jump straight to a model. So now, here is our path: six steps.

2 to 3: So now, let's start with step one: the data, and how we made the test fair.

3 to 4: Our data is ready and our test is fair. So now, step two: our first model.

4 to 5: We saw that our MLP is overfitting. So now, step three: we try to fix this overfitting.

5 to 6: We saw that the overfitting is under control. But our MLP still sees colours, not shapes. So now, step four: we need a network that sees shapes.

6 to 7: We saw how ResNet18 works. So now, let's see what it gives on our paintings.

7 to 8: But to answer the director, we also need a challenger without any neural network. So now, step five: our classical challenger, then the final test.

8 to 9: We saw how our challenger works. So now, the final test: every model, on 220 paintings never seen before.

9 to 10: We saw that the test and the validation disagree. So now, we check if the fine-tuned win was luck, with 220 new paintings.

10 to 11: Accuracy is only half of the director's question. So now, step six: what does it cost on a kiosk?

11 to 12: We saw that it is accurate, fast and cheap. So now, we have everything we need to answer the director.

Slide 1 · Is a neural network worth it on a museum kiosk?

0:00 to 0:40 · 40 s · 84 words

Good morning. Our museum wants kiosks that recognise the style of a painting from a photo.

We worked on five styles: Baroque, Cubism, Impressionism, Pop Art and Ukiyo-e. *[point at the five paintings]*

The kiosks have small CPUs, the team is small and the budget is limited. So the director asks one question: **is a neural network worth it in production, or would a simpler classical model do the job?**

> **TRANSITION TO SLIDE 2**
> To answer it, we did not jump straight to a model. **So now, here is our path: six steps.**

Slide 2 · Six steps to answer the director

0:40 to 1:10 · 30 s · 68 words

[point at the six steps, one by one] Step one: the data, and a fair test. Step two: our own neural network. Step three: we monitor its training and fix it. Step four: a pretrained ResNet18. Step five: the final tests, against a classical model. Step six: the cost on a kiosk.

Each step told us what to try next.

> **TRANSITION TO SLIDE 3**
> **So now, let's start with step one: the data, and how we made the test fair.**

Slide 3 · A fair test: one painter, one part

1:10 to 1:55 · 45 s · 101 words

We took 1 500 paintings from WikiArt, 300 per style.

We cut them into three piles: 1 061 to train, 219 to validate, that is, to choose our settings, and 220 to test, only once, at the end.

One rule matters a lot: **all the paintings of a known painter go to the same pile.** *[point at the zero]* With a random split, 23 of our 32 known painters would be in two piles, and a model could learn to recognise Monet instead of Impressionism. With our split: zero.

> **TRANSITION TO SLIDE 4**
> Our data is ready and our test is fair. **So now, step two: our first model.**

Slide 4 · Our MLP learns the paintings by heart

1:55 to 2:50 · 55 s · 116 words

Our first model is the simplest neural network of the course: a multi-layer perceptron built from nn.Linear layers and ReLU.

Before trusting it, we ran two checks. First, a model that knows nothing should have a loss of ln 5, about 1.61. *(optional) Ours started at 1.99, too confident, so we made its last layer smaller and got 1.59.* Second, it must be able to learn one small batch by heart: it does, 100 percent.

[point at the curves] The training accuracy goes to 99.7 percent, but the validation accuracy stays around 40 percent. **This is overfitting: the network learns the paintings by heart.**

> **TRANSITION TO SLIDE 5**
> We saw that our MLP is overfitting. **So now, step three: we try to fix this overfitting.**

Slide 5 · Healthy training, but the MLP overfits

2:50 to 3:45 · 55 s · 120 words

Before fixing it, we looked inside the network to check that the training itself was healthy: the gradient reaches every layer and only 4 percent of the neurons are dead. So the problem really is overfitting.

Then we attacked it with the tools of session 6. *[point at the chart]* Together, weight decay, dropout, batch norm, flips and early stopping take the MLP from 38.8 to 45.5 percent, and the gap from 0.61 to 0.12.

But one number shows a deeper limit: **the MLP finds only 7 percent of the Cubist paintings. It sees colours, not shapes.**

> **TRANSITION TO SLIDE 6**
> We saw that the overfitting is under control. But our MLP still sees colours, not shapes.
> **So now, step four: we need a network that sees shapes.**

Slide 6 · ResNet18: a network made for images

3:45 to 4:45 · 60 s · 134 words

There is a type of network made for images, which recognises shapes: the convolutional network. We used ResNet18, the smallest ResNet.

[point at the dark box] Its 18 layers slide small filters over the picture: the first layers find edges and colours, the next ones textures, the last ones shapes. *(optional) Our MLP saw 12 288 separate pixels.*

ResNet18 already learned this on 1.2 million ImageNet photos, so **first, we keep it frozen.** *[point at the 512 numbers]* It turns each painting into 512 numbers. *[point at the blue box]* We replace its last layer, made for ImageNet, by our own: from these 512 numbers to our 5 styles, 2 565 weights, the only ones we train in this first version. **Like hiring an experienced art expert, and only teaching them our five style names.**

> **TRANSITION TO SLIDE 7**
> We saw how ResNet18 works. **So now, let's see what it gives on our paintings.**

Slide 7 · A pretrained ResNet18 sees shapes

4:45 to 5:45 · 60 s · 126 words

[point at the blue bar] Training this last layer takes less than a second, and gives **88.1 percent on validation, almost twice our best MLP.** *[point at the top card]* And it does see shapes: it finds 71 percent of the Cubist paintings, against 7 for our MLP.

We also tried fine-tuning: we unfreeze its last block, 8.4 million weights instead of 2 565, so that it also learns from our paintings. On validation it looked less convincing, 85.4 percent, so we chose the frozen version. **Keep this in mind.**

[point at the bottom card] We monitored it like the MLP: the gap between train and validation is only 0.09, against 0.61 for our first MLP.

> **TRANSITION TO SLIDE 8**
> But to answer the director, we also need a challenger without any neural network. **So now, step five: our classical challenger, then the final test.**

Slide 8 · Our classical challenger: a random forest

5:45 to 6:35 · 50 s · 112 words

A classical model cannot read raw pixels well. So we first describe each painting with 102 hand-made numbers: *[point at the second box]* 82 for its colours, and 20 for the direction and strength of its brush strokes.

[point at the tree] Then a random forest: 500 decision trees. Each tree asks yes or no questions on these numbers, like "many dark colours?", until it names a style. Each one learns from a random draw of the paintings, so they make different mistakes. **They vote, and the vote cancels the mistakes.**

[point at the right card] *(optional) Like every model, we chose its settings on validation: 63.5 percent.*

> **TRANSITION TO SLIDE 9**
> We saw how our challenger works. **So now, the final test: every model, on 220 paintings never seen before.**

Slide 9 · Final test: the pretrained network wins

6:35 to 7:20 · 45 s · 101 words

We opened the 220 test paintings only once, for every model. *[point at the top bars]* The pretrained models are at the top, from 83 to 88 percent. The random forest reaches 67; our MLP 41.

Is the gap real? *[point at the blue card]* A paired bootstrap gives our frozen model a lead of **16 points over the random forest, between 9 and 24: it is not luck.**

[point at the grey card] But here is the surprise: the fine-tuned model is first on the test, while it was second on validation.

> **TRANSITION TO SLIDE 10**
> We saw that the test and the validation disagree. **So now, we check if the fine-tuned win was luck, with 220 new paintings.**

Slide 10 · A second test: the fine-tuned win was not luck

7:20 to 8:00 · 40 s · 90 words

We drew 220 new paintings from parts of WikiArt we had never opened, by painters our models never saw, and tested the same models once more.

[point at the blue bars] Same ranking: fine-tuned 87.7 percent, frozen 84.1, random forest 64, our MLP 42. *[point at the blue card]* On the 440 test paintings together, fine-tuning wins by **5 points, between 2 and 8: it is not luck.**

(optional) Our 219 validation paintings were simply too few to separate the two versions.

> **TRANSITION TO SLIDE 11**
> Accuracy is only half of the director's question. **So now, step six: what does it cost on a kiosk?**

Slide 11 · Fast and cheap enough for a kiosk

8:00 to 8:40 · 40 s · 86 words

We timed one photo at a time, from the image to the answer. *[point at the blue card]* The fine-tuned ResNet needs **53 milliseconds on one CPU thread**, 27 with four threads: faster than a blink. Even on a kiosk four times slower, it answers in a fifth of a second.

The model file is 45 megabytes, and training takes two and a half minutes on a laptop, without a GPU.

> **TRANSITION TO SLIDE 12**
> We saw that it is accurate, fast and cheap. **So now, we have everything we need to answer the director.**

Slide 12 · Our choice: a fine-tuned ResNet18

8:40 to 9:35 · 55 s · 118 words

[point at the line under the title] On the 440 test paintings, our own MLP reached 42 percent, the random forest 65, and the fine-tuned ResNet18 88.

So, our answer to the director: yes, a neural network is worth it, but a pretrained one that we reuse. **We recommend ResNet18, pretrained on ImageNet, with its last block fine-tuned on our paintings.**

[point at the left column] Why? It is accurate: one visitor in eight gets a wrong style, against one in three. It is fast enough for a modest CPU, and cheap for a small team: retrained in a few minutes.

[point at the right column] Before the launch, we must test real photos taken at the kiosks, add more styles, and measure the real hardware.

Thank you. We are happy to take your questions.